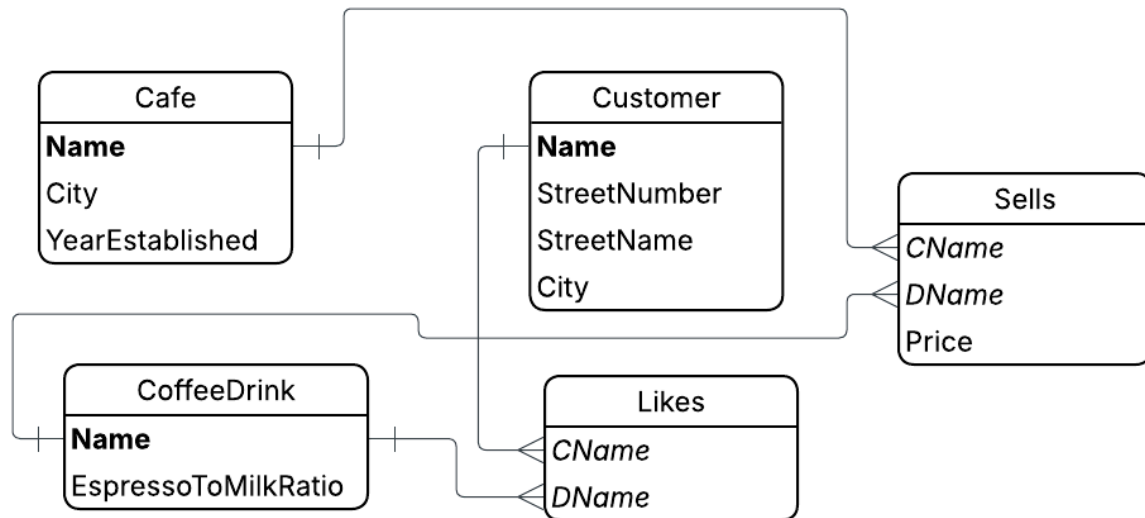


Assignment 3 Report

Task 1

Part 1



The different boxes show each table in the database, and lines from fields in one table to fields in other tables serve as foreign keys. Bolded fields are primary keys, and italicized fields are composite keys.

Part 2

I'll analyze this SQL database schema for you.

This database models a cafe/coffee shop system with five interconnected tables:

Core Entity Tables:

1. **Cafe** - Stores information about coffee shops
 - Name and City (both required)
 - YearEstablished (optional)
2. **Customer** - Tracks customer information
 - Name (required)
 - Street address components (StreetNumber, StreetName) and City (all optional)
3. **CoffeeDrink** - Catalogs different coffee beverages
 - Name (required)
 - EspressoMilkRatio (optional, stored as text like "1:2")

Relationship Tables:

4. **Likes** - Records which customers like which drinks (many-to-many relationship)
 - CName: Customer name
 - DName: Drink name
5. **Sells** - Tracks which cafes sell which drinks and at what price (many-to-many relationship)
 - CName: Cafe name
 - DName: Drink name
 - Price: Cost of the drink (up to 999.99)

The AI (Claude Sonnet 4.5) did a very good job at identifying field names, what they might potentially stand for, and even got the same relationships between tables that I did, but did not identify any foreign keys (in the actual sql file, there were no explicit definitions of any fields as foreign keys). In summary, the AI model was able to perfectly understand the database schema.

Task 2

Part 1

A)

- a) Command: select customername, streetnumber, streetname from customer where streetnumber < 100

	CUSTOMERNAME	STREETNUMBER	STREETNAME
1	Carla Mendes	88	Mission St
2	Henry Adams	44	Pennsylvania Ave
3	Isabella Cruz	77	Maple St

b)

B)

a) Command: select avg(2026 - yearestablished) as avg_age from cafe

```

      AVG_AGE
-----
      17.125

```

b)

C)

a) select cafename, avg(price) as price_USD, avg(price) * 90.56 as price_INR from sells group by cafename

CAFENAME	PRICE_USD	PRICE_INR
Bean and Barrel	6	543.36
Blue Heron Cafe	6.16666667	558.453333
Caffe Roma	6.375	577.32
Capital Grounds	5.4	489.024
Fog City Coffee	5.1	461.856
Harbor Brew	6	543.36
Morning Tide	5.75	520.72
Sunrise Roasters	5.95	538.832

b)

D)

a) select customername, city from customer where city in (select city from cafe where cafename = 'Blue Heron Cafe')

CUSTOMERNAME	CITY
Alice Johnson	Seattle
Isabella Cruz	Seattle

b)

E)

a) select distinct cafename from sells where drinkname = any (select drinkname from likes where customername = 'Alice Johnson');

```

CAFENAME
-----
Bean and Barrel
Blue Heron Cafe
Caffe Roma

```

b)

F)

a) select cafename, drinkname from sells s where exists(select 1 from likes l where l.customername = 'Grace Patel' and l.drinkname = s.drinkname)

CAFENAME	DRINKNAME
Bean and Barrel	Latte
Blue Heron Cafe	Cortado
Blue Heron Cafe	Latte
Morning Tide	Cortado

b)

G)

- a) select customer.customername, cafe.cafename, customer.city from customer inner join cafe on customer.city = cafe.city

CUSTOMERNAME	CAFENAME	CITY
Alice Johnson	Blue Heron Cafe	Seattle
Brian Lee	Harbor Brew	Baltimore
Carla Mendes	Caffe Roma	San Francisco
Carla Mendes	Fog City Coffee	San Francisco
David Kim	Morning Tide	Boston
Elena Rossi	Bean and Barrel	Chicago
Frank Turner	Sunrise Roasters	Austin
Grace Patel	Caffe Roma	San Francisco
Grace Patel	Fog City Coffee	San Francisco
Henry Adams	Capital Grounds	Washington
Isabella Cruz	Blue Heron Cafe	Seattle

b)

CUSTOMERNAME	CAFENAME	CITY
Jack Nguyen	Harbor Brew	Baltimore

H)

- a) select customer.customername, cafe.cafename, likes.drinkname from customer join likes on customer.customername = likes.customername join sells on likes.drinkname = sells.drinkname join cafe on sells.cafename = cafe.cafename

CUSTOMERNAME	CAFENAME	DRINKNAME
Alice Johnson	Caffe Roma	Cappuccino
Alice Johnson	Caffe Roma	Flat White
Alice Johnson	Bean and Barrel	Latte
Alice Johnson	Blue Heron Cafe	Latte
Brian Lee	Blue Heron Cafe	Espresso
Brian Lee	Fog City Coffee	Espresso
Carla Mendes	Caffe Roma	Flat White
David Kim	Capital Grounds	Americano
David Kim	Harbor Brew	Americano
Elena Rossi	Harbor Brew	Mocha
Frank Turner	Blue Heron Cafe	Cortado
Frank Turner	Morning Tide	Cortado
Grace Patel	Blue Heron Cafe	Cortado
Grace Patel	Morning Tide	Cortado
Grace Patel	Bean and Barrel	Latte
Grace Patel	Blue Heron Cafe	Latte
Henry Adams	Blue Heron Cafe	Espresso
Henry Adams	Fog City Coffee	Espresso
Isabella Cruz	Caffe Roma	Cappuccino
Jack Nguyen	Sunrise Roasters	Macchiato

b)

I)

- a) select customer.customername, cafe.cafename, likes.drinkname from customer join likes on customer.customername = likes.customername join sells on likes.drinkname = sells.drinkname join cafe on sells.cafename = cafe.cafename and customer.city = cafe.city


```
select customer.customername
from customer
join likes on customer.customername = likes.customername
join sells on likes.drinkname = sells.drinkname
where sells.cafename = 'Harbor Brew'
```

And the result is this:

	CUSTOMERNAME
1	David Kim
2	Elena Rossi

I gave both the part 1 and part 2 sql scripts to my selected AI model (Claude Sonnet 4.5), with the following prompt:

now that you've read through this sql file, which populates (and gives extra clarity to more ambiguous column names), write a sql query which will show customers who like ONLY drinks that are served at Harbor Brew.

It's response is as follows:

```
SELECT cu.CustomerName
FROM Customer cu
WHERE NOT EXISTS (
  SELECT 1
  FROM Likes l
  WHERE l.CustomerName = cu.CustomerName
  AND l.DrinkName NOT IN (
    SELECT s.DrinkName
    FROM Sells s
    WHERE s.CafeName = 'Harbor Brew'
  )
);
```

How this works:

1. For each customer, the subquery checks if there exists any drink they like that is NOT sold at Harbor Brew
2. **NOT EXISTS** means we only return customers where NO such drink exists
3. In other words: all drinks they like ARE sold at Harbor Brew

I think the AI model's solution makes more sense because it goes through each customer, looks at what drinks they like, sees where those drinks are sold, and only accepts those that are sold at Harbor Brew.